

# Attention Is All You Need

## Abstract

We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train.

## Introduction

Recurrent neural networks, long short-term memory and gated recurrent neural networks in particular, have been firmly established as state-of-the-art approaches in sequence modeling and transduction problems such as language modeling and machine translation.

## Multi-Head Attention

Instead of performing a single attention function with  $d_{\text{model}}$ -dimensional keys, values, and queries, we found it beneficial to linearly project the queries, keys, and values  $h$  times with different, learned linear projections to  $d_k$ ,  $d_k$ , and  $d_v$  dimensions respectively.

## Results

On the WMT 2014 English-to-German translation task, the big transformer model outperforms the best previously reported models including ensembles by more than 2.0 BLEU, establishing a new state-of-the-art BLEU score of 28.4.